Capgemini Week-11 Assessment

Assessment Question

Scenario:
You are tasked with designing and implementing an **E-Commerce Order Management System** for a retail company. The system should allow customers to browse products, place orders, and track shipments. Admins should be able to manage products, categories, and customer orders.

Requirements

1. **Entities & Relationships**
   - A **Customer** can place multiple **Orders** (One-to-Many).
   - An **Order** can contain multiple **OrderItems** (One-to-Many).
   - A **Product** can appear in multiple **OrderItems** (Many-to-Many via OrderItem).
   - A **Category** can contain multiple **Products** (One-to-Many).
   - Each **Order** has one **ShippingDetail** (One-to-One).
2. **Database Design (Code-First)**
   - Create models for `Customer`, `Category`, `Product`, `Order`, `OrderItem`, and `ShippingDetail`.
   - Use **Data Annotations** for validation (e.g., `[Required]`, `[StringLength]`, `[Range]`).
   - Configure relationships using **Fluent API** in `DbContext`.
3. **Controllers & Views**
   - Implement CRUD operations for Customers, Products, Categories, and Orders.
   - Use **Bootstrap** to style product listing pages, order forms, and tables.
   - Display a customer's order history with product details and shipping status.
   - Display admin dashboards for managing products and pending shipments.
4. **Bonus Requirements**
   - Implement a **filter** to show products by category.
   - Create a **dashboard view** showing:
     - Top 5 selling products.
     - Orders pending shipment.
     - Customer order history summary.
   - Use **ViewModels** to combine `Order + OrderItems + ShippingDetail` for display.
   - Add **authorization** so only admins can manage products and shipping.

Assessment Tasks

1. **Design models** with proper relationships and annotations.

2. **Configure DbContext** with Fluent API for many-to-many and one-to-one relationships.
3. **Implement controllers** for Customers, Products, Orders, and Admin.
4. **Build Bootstrap-styled views** for product listing, order placement, and dashboards.
5. **Create ViewModels** to display combined order and shipping data.
6. **Add filtering and authorization features** to enhance usability and security.

NOTE:- using vs-code and with empty template.